

Short, Precise, and Valid: A Computerized Adaptive Approach to Measuring the Big Five

Blinded for Review

Abstract

Abbreviated personality measures are practically important for personality research but sacrifice measurement precision, particularly at trait extremes, attenuating criterion validity in ways researchers often underappreciate. Computerized Adaptive Tests (CATs) address this by selecting items tailored to each respondent's trait level to create more accurate measures, but uptake of these methods has been limited by a lack of thorough validation and accessible implementation tools. We introduce and validate the Big Five Adaptive Inventory (BF-AI), a freely available CAT built on the 100-item IPIP-NEO via the open-source `catSurv` R package. In a simulation study, the BF-AI outperforms fixed short forms (TIPI, BFI-2-XS, Mini-IPIP) in measurement precision across the full trait distribution, with just five adaptive items matching the accuracy of the full IPIP-50. In a criterion validity study ($n = 3,029$), the BF-AI predicted seven consequential life outcomes as well as or better than fixed short forms, frequently replicating full-scale effect sizes, demonstrating that adaptive assessment can substantially reduce the brevity–precision trade-off without increasing respondent burden.

1. Introduction

The widespread adoption of short personality measures has been transformative for the field of personality research, enabling scholars to include personality questionnaires in studies where previously they might not have measured personality at all due to time and resource constraints.¹ However, this focus on brevity over comprehensiveness can have significant negative consequences. Most critically, short measures sacrifice bandwidth for fidelity, resulting in substantially lower criterion validities compared to longer assessments (Credé, Harms, Niehorster, & Gaye-Valentine, 2012).

What makes this especially striking is that the widespread shift to digital administration has created precisely the conditions under which this trade-off could be overcome, or at least ameliorated. Over the past two decades, personality assessment has migrated almost entirely to computers and handheld devices—yet this technological revolution has gone largely unexploited for personality assessment. Digital platforms are capable of moving beyond the traditional model in which all participants answer identical questions, instead tailoring items to each individual in real time. This capacity for personalized assessment could, in principle, deliver both the brevity researchers need and the measurement precision they have been forced to sacrifice.

Indeed, Computerized Adaptive Tests (CATs) are exactly the tool needed to leverage digital platforms for short yet precise personality assessment. Rather than administering a fixed set of items to every respondent, CATs draw from a larger calibrated pool, selecting at each step the item that maximizes measurement information given the individual's previous responses. Yet despite their clear potential and extensive theoretical development, CAT-based personality measures have seen limited adoption—in part, we argue, because rigorous empirical evaluations demonstrating their advantages over established short forms have been lacking.

The present study addresses this gap by introducing and validating the Big Five Adaptive Inventory (BF-AI), a freely available CAT based on the IPIP-NEO item pool (Goldberg et al., 2006).

¹Neither study reported in this article was pre-registered. All data, analysis code, and replication materials will be made publicly available at the Harvard Dataverse upon publication.

We test its predictive validity using two to five items per trait and compare performance against established short measures of personality, with the BF-AI serving as a concrete illustration of the broader potential of adaptive assessment—an approach applicable in principle to any construct for which a calibrated item bank can be assembled.

1.1. Adaptive Inventories and Personality Assessment

Traditional “paper and pencil” personality assessments follow a standardized approach where every test-taker completes the exact same test. This uniformity is sensible given that items are carefully selected and validated by test developers to capture specified latent traits. However, this approach is fundamentally inefficient because it requires everyone to respond to every item, even when some items provide no novel information about a person’s latent standing. The problem is that having participants complete items that offer redundant information wastes both participant and researcher time and resources.

The attempt to avoid this waste has a long history. As early as the 1900s, Binet employed an adaptive measurement approach allowing different individuals to respond to different questions according to their abilities. For example, it would be inefficient to ask a 7-year-old mathematics questions at the 10-year-old level if they already cannot answer 8-year-old level questions. The same logic applies directly to personality measurement: asking someone who has already demonstrated very low conscientiousness to respond to an item that only distinguishes between average and high conscientiousness levels represents an equivalent waste of time and resources.

Computerized Adaptive Tests (CATs) formalize this intuition into a rigorous psychometric framework. Rather than administering a fixed predetermined set of items, CATs select from a larger calibrated item pool, choosing at each step the item that maximizes measurement information for that particular individual. Consequently, each participant may receive a different set of items to assess the same latent trait, creating more precise individual-level measures. This approach is already well established in educational (e.g., GRE) and health assessment (e.g., PROMIS) contexts, where it consistently delivers superior measurement precision with fewer items.

Several algorithmic implementations of this framework exist in the literature (e.g., Waller & Reise, 1989; Weiss & Vale, 1987). The present paper employs one well-validated approach: a graded response model (Samejima, 1969) that characterizes each item by its discrimination and threshold parameters, uses Expected A Posteriori (EAP) estimation to update trait estimates after each response, and selects subsequent items via Minimum Expected Posterior Variance (MEPV) (van der Linden, 1998). This is one of several principled implementations available to researchers from the extensive CAT literature — a more technical description is provided in Author et al. (2017).

Despite this clear potential and well developed theoretical foundation, few CAT-based personality measures have achieved widespread use. Among the most notable are the Navy Computer Adaptive Personality Scales (NCAPS; Houston, Borman, Farmer, & Bearden, 2006) and the Tailored Adaptive Personality Assessment System (TAPAS; Drasgow et al., 2012), both of which demonstrate efficient and valid personality assessment. However, both were developed by the armed forces and their item content is restricted, preventing broader adoption. A more recent CAT built on the NEO-FFI (Nieto, Abad, & Olea, 2018) further demonstrates the utility of adaptive personality assessment, yet uptake of these methods across the field has remained limited.

The case of the CAT-based personality disorder measure (CAT-PD; Simms et al., 2011) is instructive. Developed from the ground up with the explicit goal of free distribution, the CAT-PD demonstrated clear advantages over fixed-length administration—yet researchers and clinicians preferred the fixed-length version even when administering items digitally, and the CAT version is no longer maintained. This illustrates that demonstrated superiority is not sufficient for adoption.

What has changed, however, is the technological landscape. The near-universal shift to digital data collection means that the infrastructure needed to support adaptive administration already exists across virtually all contemporary survey platforms. The technical barrier that once made CAT deployment prohibitive—the difficulty of real-time scoring and item selection—has effectively been removed by faster computers paired with open source efficient software. The opportunity to finally leverage digital platforms for short yet precise personality assessment is now within reach.

What has remained lacking is rigorous empirical validation demonstrating that adaptive admin-

istration is worth the additional effort in personality research. The present study addresses this issue directly. We introduce and validate the Big Five Adaptive Inventory (BF-AI), a freely available CAT based on the IPIP-NEO item pool (Goldberg et al., 2006), implemented through the `catSurv` R package (Montgomery & Rossiter, 2017), which integrates directly with standard web-enabled survey software such as Qualtrics. We test its predictive validity using two to five items per trait and compare performance against established short measures of personality. In doing so, the BF-AI serves as a concrete illustration of the broader potential of adaptive assessment—`catSurv` already provides validated adaptive inventories for a range of constructs including Narcissism, Schwartz values, Machiavellianism, Right-Wing Authoritarianism, Need to Evaluate, Need for Cognition, the Empathizing and Systemizing Quotients, and Social Dominance Orientation. Further, the approach is applicable in principle to any construct for which a calibrated item bank can be assembled.

1.2. Abbreviated Measures of Personality

Typical comprehensive personality inventories can exceed 200 items in length. The NEO-PI-R contains 240 items, the MPQ includes 276 items, the CPI encompasses 434 items, and the SPI contains 696 items. The more commonly used mid-size inventories are more manageable, such as the 100-item IPIP-NEO, 50-item IPIP, 60-item BFI-2, or 44-item BFI. Abbreviated personality measures, in contrast, typically contain 30 or fewer items—in the case of measuring Big Five personality traits, meaning each trait is assessed with six or fewer items. Researchers must continually balance thoroughness against time constraints, resource limitations, and participant fatigue. Consequently, most personality research employs mid-length to abbreviated scales rather than comprehensive inventories.

Abbreviated measures offer several positive attributes. Most importantly, they require minimal administration time, thereby increasing usage rates and minimizing implementation costs, allowing more constructs to be covered in the confined time range. Beyond practical considerations, they benefit participants by reducing response burden and avoiding fatigue that could potentially compromise data quality. Indeed, abbreviated questionnaires do not necessarily suffer psychometrically.

Abbreviated measures compared to medium-length alternatives do not show substantial decrements in validity (Thalmayer, Saucier, & Eigenhuis, 2011). Similarly, test-retest reliability and convergent validity remain acceptable for abbreviated scales.

While abbreviated measures demonstrate some acceptable psychometric properties, they do suffer from important limitations. Abbreviated measures necessarily reduce the information that could be captured by the set of items. This reduction is problematic for several reasons.

First, when researchers test whether other individual differences are unique or separate from “personality,” this becomes easier to demonstrate with an abbreviated measure of personality due to the limited scope of abbreviated personality measures. For example, job satisfaction shows incremental validity in predicting task performance over personality traits when short personality measures are used, but this incremental validity disappears when longer personality measures are employed (Credé et al., 2012). However, this may not necessarily indicate true incremental validity, but rather an artifact of construct underrepresentation in the abbreviated personality measure.

Second, abbreviated measures, by definition, lack breadth of content coverage. This limitation is particularly consequential given that the Big Five domains are aptly named; their conceptual breadth is substantially multifaceted. An abbreviated measure, by necessity, can only sample a fraction of these facets. Consequently, traits labeled as “conscientiousness” on comprehensive personality measures may not represent the same latent construct captured by abbreviated conscientiousness measures, as they may lose facet-level information. While content differences exist across all personality measures, these discrepancies become pronounced when only a handful of items are used. This construct underrepresentation can severely impact predictive utility, especially when an outcome is primarily driven by a specific facet that is omitted from the abbreviated scale.

Some scales attempt to counteract this lack of breadth through item construction, but this can introduce other psychometric problems. The Ten-Item Personality Inventory (TIPI; Gosling, Rentfrow, & Swann, 2003), for instance, uses adjective pairs for its items (e.g., “critical, quarrelsome” to assess (dis)Agreeableness). While this is intended to broaden content coverage within a single item, it often results in a double-barreled stimulus. The traits paired together are not synonymous.

One can be internally critical without being externally quarrelsome, or one can be quarrelsome for sport without being particularly critical. A respondent is left unsure of how to respond if they endorse one adjective but not the other, introducing ambiguity and measurement error.

Third, short inventories (and most long) are created without regard to where on the latent trait items provide information. In contrast to long or mid-length inventories, which are built from the ground up, short forms are created by selecting items from a parent inventory that exhibit the highest factor loadings. While seemingly rigorous, this method preferentially selects “prototypical” items that are highly similar in content and provide maximum measurement information near the center of the trait distribution. The result is a psychometrically redundant scale, offering a very narrow assessment of the trait, and one that is most precise for average individuals but highly imprecise for those with high or low trait levels.

Ideally, an abbreviated measure would select items that provide complementary, not redundant, information to ensure both content breadth and uniform precision across the trait continuum. Some thoughtful attempts have been made, such as in the creation of the Mini-IPIP (Donnellan, Oswald, Baird, & Lucas, 2006), where the authors aimed to preserve facet representation from the original NEO inventory. However, without a formal psychometric framework like Item Response Theory to guide item selection, there was no explicit mechanism to guarantee that the chosen items would provide information across a wide range of the latent trait. This leaves a critical gap for a methodology that can achieve brevity without sacrificing content validity or equitable measurement precision across trait levels.

These shortcomings, inherent to any fixed-length abbreviated measure, can theoretically be overcome by a CAT-based approach to personality assessment. Unlike a static short form, a CAT leverages a large, pre-calibrated item bank from which an algorithm dynamically selects a unique set of items for each individual. In principle, this methodology directly addresses the primary weaknesses of traditional short forms. The large item pool provides the necessary content breadth to avoid construct underrepresentation, while the adaptive algorithm is designed to select items

that provide maximum information at the individual's specific trait level.² The theoretical result is a measurement instrument that is both highly efficient and psychometrically robust, providing equitable precision across the entire trait continuum.

However, these are largely theoretical claims in the context of the Big Five assessment. While the principles of CAT are well-established, a systematic empirical demonstration of its advantages in measurement precision and criterion validity against a comprehensive suite of modern abbreviated and full-scale personality inventories is currently lacking in the literature.

1.3. Predictive Validity of Personality

Personality traits are excellent predictors of consequential life outcomes, including mortality and divorce, often decades in advance. Their predictive power remains significant even when controlling for a host of demographic and background variables, holds across different raters, and is comparable to or superior to established predictors like cognitive ability and socioeconomic status. Moreover, these associations are highly replicable and generalizable, holding across demographic groups (e.g., gender, age, SES) and national contexts (Beck & Jackson, 2022; Soto, 2019).

However, the criterion validity of *abbreviated* personality scales remains under-examined and a pressing issue. The predictive power of personality is likely attenuated or biased when assessed with abbreviated scales, which limit multiple forms of validity (e.g., convergent, discriminant, criterion). This attenuation is a direct consequence of their typical construction process, which prioritizes internal consistency (fidelity) by selecting a few highly correlated items. This results in a narrow operationalization of the broad trait, leading to construct deficiency—a failure to capture the full conceptual scope of the construct (Messick, 1995). As a consequence, criterion-related validity is systematically reduced; indeed, meta-analytic evidence confirms that short measures exhibit weaker predictive validity than their longer counterparts (Credé et al., 2012).

This deficiency can also paradoxically increase the risk of Type I errors in incremental validity studies, where another variable may appear to predict an outcome “above and beyond” the personal-

²The algorithm can also be easily adapted to choose items that cover the full spectrum of facets.

ity measure, an effect that would disappear if the full, more valid personality construct had been controlled for.

This trade-off between practicality and validity is openly acknowledged by the creators of prominent short scales. They often caution that their instruments should be used only when situational constraints, such as in large-scale panel studies or intensive longitudinal designs, make longer inventories prohibitive (e.g., Donnellan et al., 2006; Gosling et al., 2003). This necessary compromise, however, highlights a critical gap in personality assessment methodology. Despite the plea to use abbreviated measures only out of necessity, a powerful alternative that promises both brevity and psychometric robustness has not been systematically tested. Specifically, the question remains whether a CAT-based short form, by leveraging a large item bank to optimize measurement for each individual, can overcome the deficiencies of fixed-length scales and restore the predictive validity lost to abbreviation.

1.4. Current Study

The present study was designed to address the aforementioned limitations of fixed-length personality measures by introducing and rigorously validating the Big Five Adaptive Inventory (BF-AI). Our overarching goal was to provide a comprehensive evaluation of whether the BF-AI could achieve the precision closer to the long-form inventory with the brevity of an abbreviated scale, thereby resolving the trade-off between psychometric robustness and practical efficiency.

To accomplish this, our investigation proceeded in two main stages. The first stage involved a large-scale simulation study designed to provide a systematic and controlled evaluation of the BF-AI's measurement precision. We compared the BF-AI's ability to recover true latent trait scores against a comprehensive suite of benchmarks, including randomly selected items, widely used abbreviated inventories of varying lengths (e.g., TIPI, BFI-2-XS, Mini-IPIP), and full-scale measures (e.g., IPIP-50, BFI-2, IPIP-NEO). This allowed for a fine-grained analysis of the BF-AI's efficiency and its capacity to provide equitable measurement across the entire trait continuum.

The second stage sought to validate the BF-AI in an empirical context by assessing its criterion-

related validity. We administered the BF-AI alongside three fixed-length counterparts—the TIPI (2 items/trait), the BFI-2-XS (3 items/trait), and the Mini-IPIP (4 items/trait)—and examined their relative power in predicting a diverse set of 7 consequential life outcomes. To ensure a robust test, these outcomes were drawn directly from the comprehensive personality-outcome framework established in the LOOPR project (Soto, 2019). We hypothesized that the BF-AI would demonstrate superior measurement precision and that its correlations with external criteria would be comparable to those obtained with the full-length battery, thereby establishing it as a psychometrically sound and highly efficient alternative for personality assessment.

2. Method

2.1. Samples

The BF-AI development and validation process consists of two stages. First, we calibrated the item information parameters for the adaptive inventory using a hybrid sampling strategy to maximize both statistical power and demographic representativeness from two independent samples. Second, we conducted a validation study to assess the criterion validity of the defined instrument against established short-form measures of Big Five personality in predicting major life outcomes.

Sample 1. The primary calibration sample consisted of 774,410 participants who completed the 100-item International Personality Item Pool-NEO (IPIP-NEO; Goldberg et al., 2006) through the myPersonality Facebook application (Kosinski, Matz, Gosling, Popov, & Stillwell, 2015). Although the sample constitutes a convenience population with limited control over representativeness, its extraordinary size provided sufficient power to obtain highly stable IRT parameter estimates. These parameters formed the foundation for developing the adaptive inventory’s item selection algorithm.

Sample 2. To enhance the generalizability of item parameters and ensure that the adaptive algorithm performed consistently across diverse demographic groups, a supplementary calibration sample was collected through YouGov in June 2018. This sample included 1,500 U.S. adults and was designed to be quota-matched to Census benchmarks on gender, age, race, and education. By combining the large-scale convenience sample from myPersonality with a probability-based,

demographically representative sample, the calibration process balanced statistical precision with representativeness, thereby reducing the risk of bias in the adaptive algorithm's parameter estimates.

Sample 3. The criterion validation sample was recruited through Lucid Theorem, an online participant pool, in March 2021. A total of 4,136 U.S. adults were recruited; after excluding participants who failed an attention screener or did not complete the survey, 3,029 remained for analysis. Participants first completed the adaptively administered BF-AI, which selected 25 items (five per Big Five domain) from the original IPIP-NEO inventory (Goldberg et al., 2006). To provide benchmark comparisons, participants also completed three widely used short-form measures of the Big Five: the Ten-Item Personality Inventory (TIPI; Gosling et al., 2003), the Big Five Inventory-2 Extra Short Form (BFI-2-XS; Soto & John, 2017), and the Mini-IPIP (Donnellan et al., 2006). Participants then completed a set of items assessing a wide range of life outcomes, classified and implemented from Soto's (2019) large-scale replication study of personality prediction, including social and romantic relationship quality, work satisfaction, civic engagement, physical health, and behavioral outcomes such as antisocial or criminal behavior. A summary of item wording and response anchors is provided in Table 1. Complete item text, display conditions, and scoring rules are provided in Appendix A. Demographic characteristics of the final analytic sample are reported in Table 2 (Appendix B).

2.2. Measures

International Personality Item Pool-NEO (IPIP-NEO) and Big Five Adaptive Inventory (BF-AI). The item bank for the BF-AI was the IPIP-NEO (Goldberg et al., 2006), a public-domain 100-item measure of the Big Five personality domains (Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism), with 20 items per domain rated on a 5-point scale from 1 (*very inaccurate*) to 5 (*very accurate*). The BF-AI adaptively selects items from this bank based on each participant's response pattern; full details of the calibration procedure and CAT algorithm are provided in the Analytic Strategy below.

Benchmark Short-Form Measures. To evaluate the performance of the BF-AI relative to existing

brief instruments, several established personality measures were used across the two studies. In the criterion validity study, three short-form measures were administered alongside the BF-AI: the TIPI (Gosling et al., 2003), which assesses the Big Five with two items per domain on a 7-point response scale; the BFI-2-XS (Soto & John, 2017), a 15-item measure with three items per domain; and the Mini-IPIP (Donnellan et al., 2006), a 20-item short form of the IPIP that uses four items per domain and the same 5-point response scale as the full IPIP-NEO. The simulation study incorporated a broader set of benchmarks spanning a wider range of lengths, additionally including the IPIP-50 (10 items per domain), the BFI-2 (15 items per domain), and the full IPIP-NEO (20 items per domain). Together, these benchmarks enable direct comparisons of the BF-AI's performance against established instruments at multiple levels of survey brevity, from the most abbreviated short forms to comprehensive full-length inventories.

Life Outcome Measures. To assess the criterion and predictive validity of the BF-AI, participants completed a series of life outcome measures. Six criteria were drawn directly from the item batteries developed by Soto (2019) for the LOOPR project: peer acceptance, romantic relationship satisfaction, job satisfaction, religious beliefs, volunteerism, and criminal behavior. Religious beliefs were assessed using the six-item religious beliefs and behavior subscale of the LOOPR spirituality battery (items 25–30), reverse-coded so that higher scores indicate stronger religiosity. A seventh criterion, self-rated health, was included to represent an important life outcome domain not captured by the other measures. Because objective health outcomes cannot be assessed in an online survey context, participants reported their overall health using a widely used single-item self-rated health measure (“In general, would you say your health is...?”; Idler & Benyamini, 1997), which assesses perceived health status. A summary of item wording and response anchors is provided in Table 1. Complete item text, display conditions, and scoring rules are provided in Appendix A.

2.3. Analytic Strategy

The analytic strategy proceeded in three sequential stages. The first stage involved the psychometric calibration of the item bank and complete specification of the CAT algorithm. Following calibration,

Table 1: Life Outcome Measures: Item Text and Response Anchors

Construct	Item Text	Response Anchors
Health	In general, would you say your health is	1 = Poor, 2 = Fair, 3 = Good, 4 = Very good, 5 = Excellent
Peer Acceptance	In relation to people you know, how would you describe your popularity among your peers? Are you currently married? Do you currently have a romantic partner for at least one month?	1 = Extremely unpopular . . . 9 = Extremely popular 1 = Yes, 2 = No 1 = Yes, 2 = No
Romantic Relationship	To what extent do you reveal or disclose very intimate facts about yourself to your spouse or partner? How close do you feel toward your spouse or partner? To what extent do you have a sense of “belonging” with your spouse or partner? How often do you and your spouse or partner argue? (R) To what extent does your spouse or partner demand too much of your time? (R) To what extent do you try to change things about your spouse or partner that bother you? (R) The working conditions The way my co-workers get along with each other	1 = Not at all . . . 5 = Very much
Work Satisfaction	Being able to keep busy all the time The chance to work alone on the job The way my boss handles his/her workers The competence of my supervisor in making decisions	1 = Very dissatisfied . . . 5 = Very satisfied
Volunteer Behavior	Have you ever volunteered? Are you currently volunteering? Do you plan on volunteering during the next two months?	1 = No, 2 = Yes
Criminality	Theft of something worth \$50 or less Theft of something worth more than \$50 Vandalism Robbery Assault (with or without a weapon) Driving while intoxicated	1 = Never, 2 = Once or twice, 3 = Several times, 4 = Many times
Religious Beliefs	I believe that God or a Higher Power is responsible for my existence. I believe that going to religious services is important. I see God or a Higher Power present in all the things I do. I seldom, if ever, pray. (R) I have made no efforts to include religious practices as part of my life. (R) I do not see a Higher Power having any influence over my life. (R)	1 = Strongly disagree . . . 5 = Strongly agree

Note. (R) = reverse scored before averaging, so that higher composite scores indicate greater religiosity. All items except Health were implemented directly from the LOOPR project (Soto, 2019). Religious Beliefs items are the six-item religious beliefs and behavior subscale of the LOOPR Spirituality battery (items 25–30); Peer Acceptance corresponds to the single LOOPR Friendship item. Health was assessed with a standard single-item self-rated health measure (Idler & Benyamini, 1997) not included in the LOOPR battery; this item captures perceived rather than objectively measured health status.

the second stage evaluated measurement precision through a simulation-based study comparing the BF-AI against established abbreviated measures, random-subsetted measurements, and full-length measures. The final stage assessed criterion validity by examining the BF-AI's ability to predict major life outcomes relative to established short-form measures.

2.3.1. Stage 1: CAT Specification

Implementing a CAT requires five core specifications: a set of calibrated item parameters, a prior distribution over the latent trait, a trait estimation procedure, an item selection procedure, and a stopping rule that determines when to conclude administration. Many reasonable choices exist for each of these components. We describe our decisions below and refer readers to Author et al. (2017, 2022) for full technical details and additional options.

Item parameters were obtained by fitting five separate Graded Response Models (GRMs; Samejima, 1969) to the 100-item IPIP-NEO calibration data, one for each Big Five trait, using the pooled data from Calibration Samples 1 and 2. This yielded a discrimination parameter (α) for each item, indicating how effectively it differentiates among individuals along the trait continuum, and a set of threshold parameters (β) defining the trait level required to endorse each response category.

The prior distribution for each trait was centered on the sample mean from the nationally representative sample (Sample 2), with a standard deviation set to encompass the full range of trait estimates observed across both calibration samples. Trait estimation used Expected A Posteriori (EAP) estimation (Bock & Mislevy, 1982), which updates the posterior distribution of the latent trait after each response by combining this prior with the likelihood derived from the administered item's α and β parameters.

Item selection was governed by the Minimum Expected Posterior Variance (MEPV) criterion (van der Linden & Pashley, 2000). At each step, the algorithm calculates the expected posterior variance that would result from administering each remaining item given the respondent's current response profile, and selects the item expected to produce the greatest reduction in uncertainty about the respondent's true trait level.

Finally, rather than using a precision-based stopping rule, the CAT was configured to administer exactly five items per trait (25 items total). This fixed-length design facilitates direct comparisons with established short-form measures: by collecting five items per trait for every respondent, trait scores can be estimated from any subset of those items, producing BF-AI versions that precisely match the length of each fixed benchmark inventory.

2.3.2. Stage 2: Evaluation of Measurement Precision

A primary objective of the second stage was to rigorously evaluate the measurement precision of the BF-AI. A fair comparison against established abbreviated and full-length inventories is complicated by two significant methodological challenges. First, while all examined inventories purport to measure the Big Five, different inventories with various item combinations may tap into unique facets of each trait, meaning they do not perfectly represent the same underlying latent construct. Second, while a fair comparison requires a within-participant design, it is infeasible to administer all relevant inventories to each participant due to excessive burden.

To overcome these limitations, we designed a comprehensive simulation study using the simulation functionality of the `catSurv` package. This approach establishes a common metric for all batteries by ensuring all items load onto a unified latent trait scale for each Big Five dimension, thereby making all comparisons of measurement precision directly interpretable.

Procedure. The simulation proceeded in two stages (a lengthier description of this procedure is described in Appendix C). The first stage established a common latent metric across all inventories. Because the inventories were originally calibrated on different samples, their item parameters are not directly comparable—an item’s difficulty parameter from the IPIP-NEO calibration, for instance, is not on the same scale as the corresponding parameter from the BFI-2 calibration. To resolve this, we proceeded as follows: (i) initial item parameters were estimated for each inventory from its respective publicly available dataset—specifically, the IPIP-NEO from the YouGov dataset, the BFI-2 and BFI-2-XS from the LOOPR dataset (Soto, 2019), and the TIPI, IPIP-50, and Mini-IPIP from the Lucid validation dataset; (ii) a set of 1,000 simulated respondents was generated, each

assigned a true latent trait score (θ) drawn from $\text{Unif}[-4, 4]$,³ and for each respondent a complete set of responses to all items across all inventories was simulated, with response probabilities determined by the initial item parameters; (iii) all item parameters were then re-estimated jointly from this simulated dataset, placing every item onto a single common latent scale for each Big Five dimension. The result is a unified item bank in which the parameters of all items—regardless of which inventory they originally came from—are expressed in the same metric and can be directly compared.

The second stage generated the evaluation dataset used to assess measurement precision. A fresh sample of 1,000 simulated respondents was drawn from $\text{Unif}[-4, 4]$, and probabilistic item responses were generated for each using the unified item parameters from stage one. These respondents’ true θ values served as the ground truth against which the estimates recovered by each measurement approach—the BF-AI, the matched fixed-length benchmark inventory, and a random-item baseline of equivalent length—were compared.

Evaluation Conditions and Metrics. Within this simulation framework, the performance of the BF-AI was benchmarked against two comparison standards. The first consisted of random-item batteries, constructed by simple random sampling (without replacement) from the full item bank for each respondent, matched in length to the BF-AI. These served as a lower-bound baseline to quantify precision gains attributable to the adaptive algorithm versus chance. The second consisted of established fixed-length abbreviated and full-length inventories, with each matched to the corresponding BF-AI length: the TIPI (2 items/trait) at two items, the BFI-2-XS (3 items/trait) at three items, the Mini-IPIP (4 items/trait) at four items, the IPIP-50 (10 items/trait) at five items, the BFI-2 (15 items/trait) at six items, and the IPIP-NEO (20 items/trait) at seven items. Note that for lengths of five items and above, the fixed benchmark inventories are substantially longer than the BF-AI, providing a particularly demanding test of adaptive efficiency.

Measurement precision was quantified as the absolute difference between the true and estimated latent trait position ($|\theta - \hat{\theta}|$), providing a direct measure of estimation accuracy across the full range

³The latent trait scale is fixed so that the population variance of θ is 1. Under this convention, a range of $[-4, 4]$ spans four standard deviations on either side of the mean, encompassing virtually the full range of trait variation observed in practice.

of the trait distribution. This metric allows for both absolute assessments of accuracy and relative comparisons across measurement approaches.

2.3.3. Stage 3: Criterion Validity Analysis

The final stage assessed the criterion validity of the BF-AI by examining its ability to predict 7 consequential life outcomes. Trait scores for the BF-AI were estimated at four levels of brevity (2, 3, 4, and 5 items per trait). The predictive effect sizes of each version were then benchmarked directly against its fixed-length counterpart of the same length: the TIPI (2 items/trait), the BFI-2-XS (3 items/trait), and the Mini-IPIP (4 items/trait), as well as the original effect sizes reported in the LOOPR project (Soto, 2019) where available. A key methodological distinction was made in scoring: trait scores for all versions of the BF-AI were estimated using the EAP method based on the GRM parameters, whereas scores for the fixed-length benchmark inventories and the life outcome scales were calculated using traditional summed scoring.

The relationships between each estimated personality trait level and the life outcomes were assessed using Pearson’s correlation, which is scale-independent and thus provides a fair basis for comparing predictive effect sizes across measures that use different scoring metrics—including the IRT-based EAP estimates from the BF-AI and the summed scores from the fixed-length inventories.

3. Results

3.1. Measurement Precision

The simulation study was designed to evaluate the measurement precision of the BF-AI relative to baseline fixed batteries and random selection. Precision was operationalized as absolute error (i.e., bias), representing the discrepancy between simulees’ true latent trait scores (θ) and their estimated scores ($\hat{\theta}$). Figure 1 illustrates the key patterns using the Agreeableness four-item condition as a representative example before the full results across all five traits and six test lengths are presented.

For each trait value (θ), the BF-AI demonstrated substantially lower estimation bias than the random-item selection baseline across the full range of the trait distribution. The fixed-length

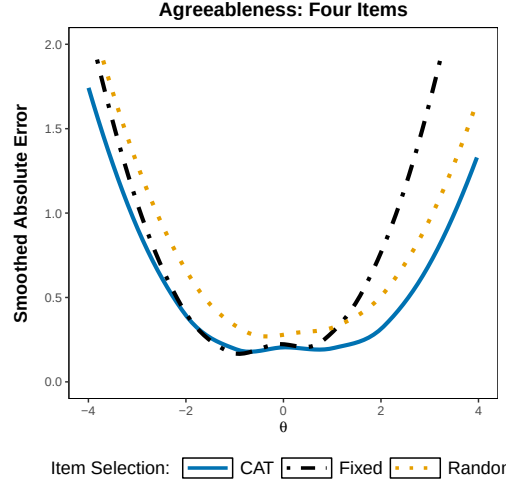


Figure 1: Bias of Estimation: Agreeableness, Four-Item Condition (Illustrative Example)

Note. Smoothed absolute error ($|\theta - \hat{\theta}|$) as a function of true trait level (θ) for the Agreeableness four-item condition. The blue solid line represents the BF-AI (adaptive selection); the black dash-dot line represents the Mini-IPIP (fixed selection); the orange dotted line represents a random-item selection baseline.

benchmark inventories, by contrast, show a different pattern: while they outperform random selection for individuals at or below the center of the distribution, they exhibit higher bias than both the BF-AI and random selection for individuals with positive trait levels. This asymmetry likely reflects the item composition of fixed short forms, which tend to include items that are more informative in the lower half of the trait distribution, providing relatively precise measurement for individuals with average or below-average trait levels while degrading for those in the upper tail. The BF-AI’s adaptive algorithm sidesteps this limitation entirely, selecting items that are maximally informative at each individual’s current trait estimate regardless of where they fall in the distribution, and producing a consistently flat and low error curve across the full range of θ

Crucially, the shape of the error curves highlights a distinct advantage of the adaptive design. The randomly selected inventory exhibited a pronounced U-shaped error function, indicating that measurement precision is highest near the mean of the trait distribution ($\theta \approx 0$) and degrades substantially toward the extremes. In contrast, the BF-AI produced a flatter, lower error curve across the full range of θ , indicating that the BF-AI provides more equitable and robust measurement, effectively covering a wider portion of the latent trait continuum with high precision. For individuals

with non-average trait levels, the BF-AI delivers more accurate estimations than a combination of randomly selected tests of the same length could achieve or even carefully chosen static short battery.

Turning to comparisons across test lengths (Figure 2), the analyses first focused on the context of extreme brevity, using measurements with lengths of two, three, and four items per trait (see rows 1–3). In general, the adaptive inventory either outperforms or performs equally to its respective fixed inventory benchmark. This advantage is especially pronounced at three items, compared to the BFI-2-XS, and four items, compared to the Mini-IPIP, where the adaptive inventory provides substantially better coverage across the full range of true θ . The one exception is the two-item setting, benchmarked against the TIPI, where differences are smaller. This is an expected outcome. With only two items, the opportunity to dynamically optimize the selection of the second item based on the response to the first is inherently limited, thus mitigating the adaptive advantage.

The evaluation was then extended to longer test lengths—five, six, and seven items—to assess the BF-AI’s efficiency relative to full-scale measurement (see rows 4–6 of Figure 2). The results demonstrate that the BF-AI’s psychometric advantage represents a fundamental gain in measurement efficiency rather than a marginal improvement. With only five items, the adaptive administration achieved a level of measurement precision comparable to the full 10-item-per-trait IPIP-50. The six-item BF-AI consistently demonstrated lower absolute error than the full 15-item-per-trait BFI-2 across the entire trait spectrum. At the seven-item level, the BF-AI achieved measurement precision broadly comparable to the full 20-item-per-trait IPIP-NEO—a inventory more than twice its length—though the fixed battery retains a modest advantage in some extreme trait regions. These findings suggest that the BF-AI can yield trait estimates that are as precise, or even more precise, than full-scale inventories while using 50–60% fewer items.

Full numerical results for all method-by-length-by-trait conditions are provided in Table 3 (Appendix D).

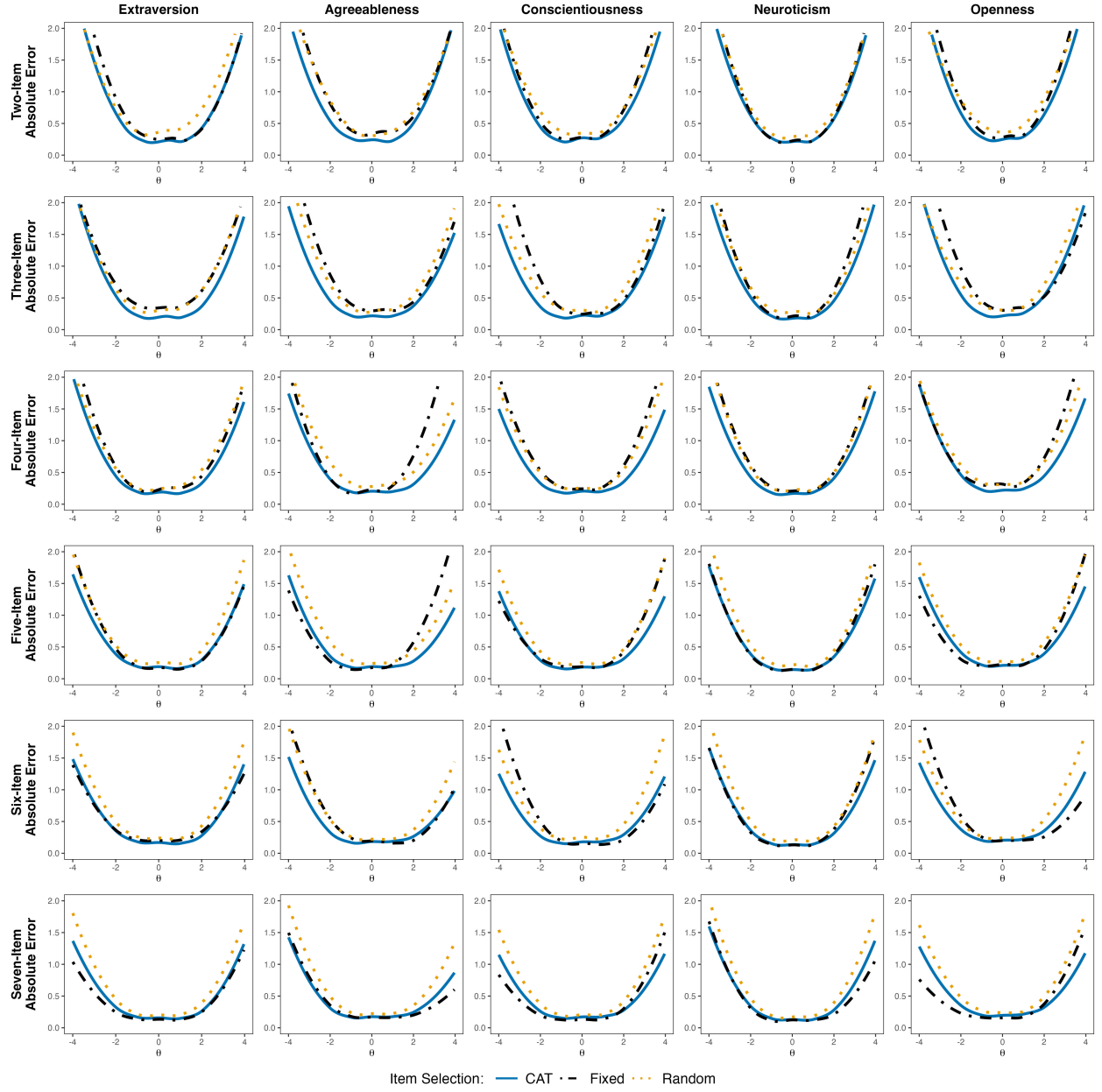


Figure 2: Bias of Estimation Based on Simulation

Note. Smoothed absolute error ($|\theta - \hat{\theta}|$) across all five traits (columns) and six test lengths (rows). Line color and type indicate the item selection method: blue solid = BF-AI (adaptive); black dash-dot = fixed benchmark inventory; orange dotted = random selection. Fixed benchmarks by row: two items (TIPI; 2 items/trait), three items (BFI-2-XS; 3 items/trait), four items (Mini-IPIP; 4 items/trait), five items (IPIP-50; 10 items/trait), six items (BFI-2; 15 items/trait), and seven items (IPIP-NEO; 20 items/trait).

3.2. Criterion Validity

The second stage of analysis assessed the criterion-related validity of the adaptive inventory by comparing its predictive power to that of established abbreviated inventories across 7 life outcomes. We first examine Criminal Behavior in detail as a clear illustration of the key patterns, then present the full results.

The BF-AI demonstrated strong validity for Criminal Behavior, replicating well-established theoretical associations with only a handful of items. As shown in Figure 3, the four-item BF-AI yields the expected directional pattern: negative correlations with Agreeableness ($r = -.25$) and Conscientiousness ($r = -.18$), a positive correlation with Neuroticism ($r = .14$), modest negative associations with Openness ($r = -.06$), and near-zero association with Extraversion ($r = -.04$). These associations are broadly consistent across both adaptive and fixed-length inventories, suggesting that even very brief measures can recover the core personality–criminality relationships.

Where the measures diverge, the BF-AI tends to align closely with the full-scale BFI-2 effect sizes from the LOOPR project (Soto, 2019), indicated by the grey reference lines in Figure 3. The BF-AI confidence intervals at all four test lengths span or closely approach the LOOPR benchmarks for trait–criminality associations. For Conscientiousness in particular, the fixed-length inventories sometimes depart further from the LOOPR reference values than the BF-AI does, consistent with our earlier discussion of how item selection bias in fixed short forms can shift criterion validity estimates.

Extending the analysis to all 7 life outcomes (Figure 4), the BF-AI generally demonstrated criterion validity comparable to fixed-length inventories of equivalent length. Where the two approaches diverge, however, the BF-AI more consistently tracks the full-scale LOOPR benchmarks. This pattern is clearest for several trait–outcome pairs: Neuroticism predicting Job Satisfaction, Agreeableness and Conscientiousness predicting Religious Beliefs, Extraversion, Agreeableness, and Conscientiousness predicting Romantic Satisfaction, Extraversion predicting Volunteerism, and Conscientiousness predicting Criminal Behavior. In each of these cases, the BF-AI confidence

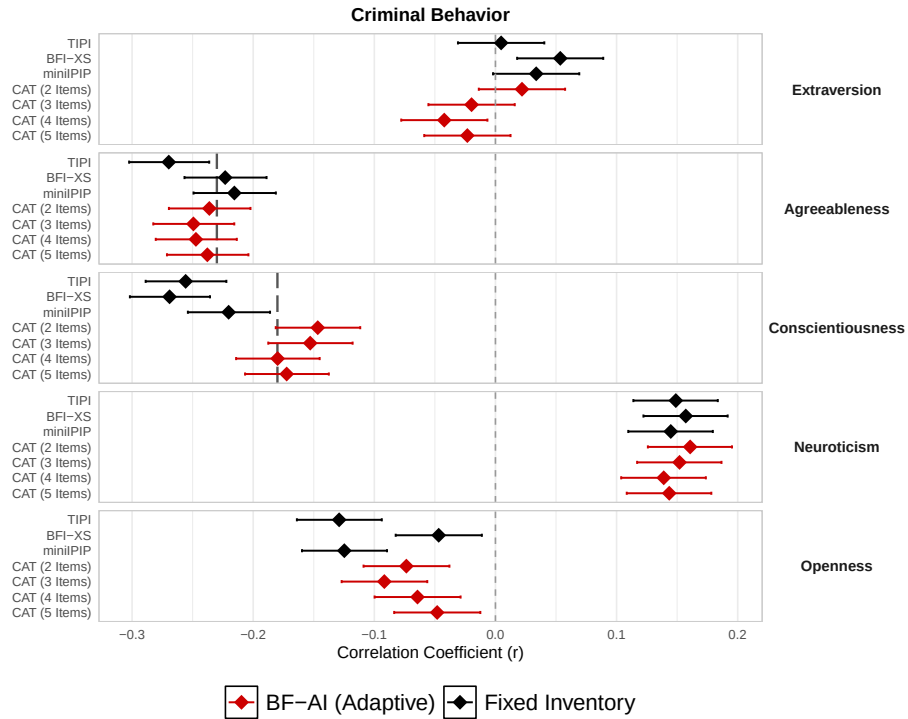


Figure 3: Criterion Validity: Criminal Behavior

Note. Correlation coefficients (r) between personality trait estimates and Criminal Behavior, with 95% confidence intervals. Red = BF-AI (adaptive); black = fixed inventories (TIPI, BFI-XS, miniIPIP). Dashed grey vertical lines indicate full-scale BFI-2 effect sizes from the LOOPR project (Soto, 2019). CAT-2 through CAT-5 refer to the BF-AI administered with two through five items per trait.

intervals more closely span or approach the LOOPR reference values than those of the matched fixed-length inventories. For three additional pairs—Agreeableness predicting Volunteerism, Neuroticism predicting Romantic Satisfaction, and Agreeableness predicting Criminal Behavior—the adaptive and fixed inventories perform similarly, with neither consistently closer to the full-scale benchmark.

The two exceptions to this overall pattern are worth noting. For Extraversion predicting Peer Acceptance and Extraversion predicting Job Satisfaction, the fixed-length inventories more closely recover the LOOPR benchmarks, with the BF-AI producing somewhat larger correlations than the reference values. This pattern is theoretically interpretable—adaptive selection may amplify certain facets of Extraversion that are particularly diagnostic for individual respondents but not uniformly predictive of these outcomes at the population level—though it warrants further investigation.

In sum, the BF-AI demonstrates criterion validity that is broadly comparable to fixed-length counterparts and, in the majority of cases, better recovers full-scale effect sizes, confirming its utility as an efficient alternative to established short forms.

Complete correlation tables for all five traits associations with all life outcomes are provided in Appendix E (Tables 4–8). A descriptive summary of the items most frequently selected by the adaptive algorithm across Sample 3 is provided in Table 9 (Appendix F).

The appendices also present several additional robustness checks based on the Sample 3 data. First, to assess convergent validity, Figure 5 (Appendix G) presents the full inter-measure correlation matrices for each trait. At matched lengths, BF-AI scores correlate moderately to strongly with their fixed-length counterparts (Table 10), ranging from $r = .48$ (Agreeableness and Openness, CAT-2 vs. TIPI) to $r = .80$ (Neuroticism, CAT-5 vs. BFI-2-XS). Neuroticism shows the highest convergent validity across all measure pairs, while Agreeableness and Openness show the lowest.

We also compared linear model fits against generalized additive models across all 35 trait–outcome pairs to verify that Pearson r (which assumes linearity) adequately characterizes these associations. Results are reported in Appendix H and confirm that all associations are well-described as linear. Finally, we consider invariant measurement across observed demographic groups; specifically considering gender, where we are best powered for this analysis. Appendix I, show that

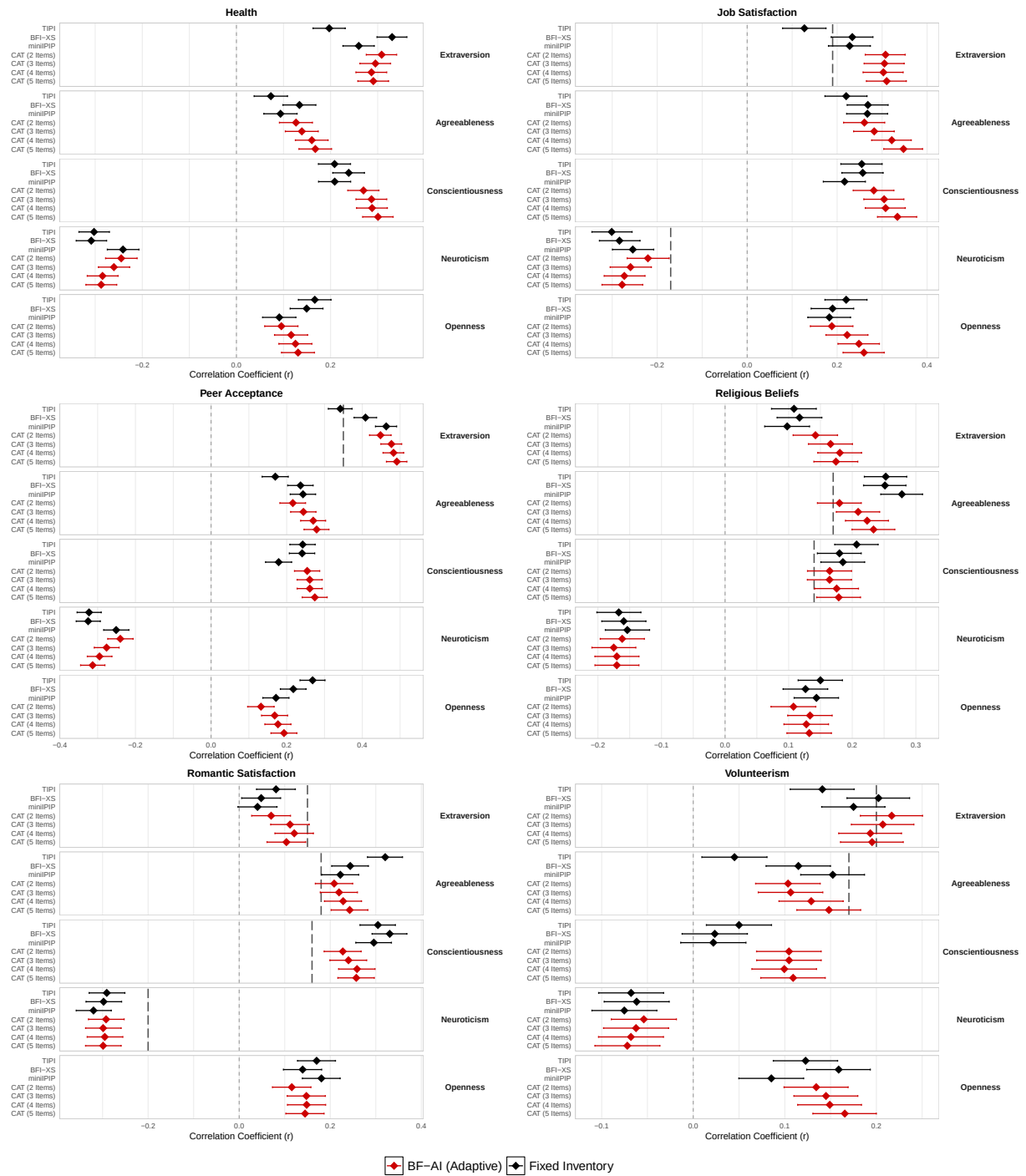


Figure 4: Criterion Validity Across Six Life Outcomes

Note. Correlation coefficients (r) with 95% confidence intervals. Red = BF-AI (adaptive); black = fixed inventories (TIPI, BFI-XS, miniIPIP). Dashed grey vertical lines indicate full-scale BFI-2 effect sizes from the LOOPR project (Soto, 2019). CAT-2 through CAT-5 refer to the BF-AI administered with two through five items per trait.

criterion validity estimates are consistent across men and women, with no systematic pattern of attenuation or enhancement for either group.

4. Discussion

4.1. Summary of Findings

The present research introduced and validated the Big Five Adaptive Inventory (BF-AI), a computerized adaptive measure of the Big Five personality traits built on the IPIP-NEO item pool. Across two studies—a large-scale simulation and an empirical criterion validity study—the BF-AI consistently demonstrated advantages over fixed-length abbreviated measures while preserving the brevity that makes short-form assessment practical.

The simulation study established that adaptive item selection produces more accurate trait estimation than fixed-length short forms, particularly for individuals at the extremes of the trait distribution. Whereas fixed short forms and randomly assembled batteries exhibited a more characteristic U-shaped bias function—performing well near the trait mean but degrading substantially toward the tails—the BF-AI produced a flatter error curve across the full range of trait values. This finding directly addresses one of the most consequential limitations of existing short forms: their failure to provide precise measurement for the individuals who deviate most from average. Beyond comparisons with abbreviated measures, the simulation revealed that the BF-AI achieves a level of measurement precision comparable to inventories two to four times its length, suggesting that adaptive administration represents a fundamental gain in psychometric efficiency rather than a marginal improvement over fixed designs.

The criterion validity study corroborated these simulation-based findings in an applied context. Across a diverse set of consequential life outcomes, the BF-AI demonstrated predictive validity that was comparable to, and in several cases exceeded, that of fixed-length inventories of equivalent length. Critically, the BF-AI better replicated the substantive effect sizes reported for full-scale inventories in the LOOPR project (Soto, 2019), suggesting that adaptive administration can recover much of the criterion validity typically sacrificed by abbreviation.

4.2. Implications for Personality Assessment

These findings carry several implications for how researchers approach personality measurement, particularly in contexts where efficiency is at a premium. The central message is that the long-assumed trade-off between brevity and validity is not an immutable feature of personality assessment; it is, in large part, a consequence of how items have traditionally been selected. By drawing on the full content of the IPIP-NEO and selecting items that are maximally informative for each individual, the BF-AI demonstrates that researchers need not choose between a test that is short and a test that is accurate.

This has immediate practical value for the many research contexts in which long personality inventories are genuinely infeasible. Large-scale panel surveys, experience-sampling studies, ecological momentary assessment protocols, and studies in which personality is one among many constructs of interest all place a premium on assessment efficiency. In such contexts, researchers have historically been forced to rely on instruments like the TIPI or Mini-IPIP with the explicit acknowledgment that doing so incurs psychometric costs (Donnellan et al., 2006; Gosling et al., 2003). The present results suggest that adaptive administration can substantially reduce those costs without increasing participant burden.

Beyond these immediate applications, the findings point toward a broader opportunity for the field. The infrastructure for adaptive assessment already exists. The widespread adoption of web-based survey platforms means that virtually all contemporary personality data collection occurs in an environment capable of supporting item-by-item adaptive administration. The technical barrier that has historically limited the use of CAT-based personality measures—the need for real-time scoring and item selection algorithms—has been effectively eliminated by tools such as the `catSurv` R package and the accompanying [URL removed for blind review] platform, which integrate directly with standard survey software. What has been lacking is not the technology but the validated instruments and the empirical demonstration that adaptive administration is worth the modest implementation effort.

The present paper addresses this gap specifically for Big Five personality assessment, but

the logic extends naturally to other personality constructs and, more broadly, to psychological assessment writ large. Adaptive administration has already demonstrated substantial advantages in related domains, including measures of narcissism, authoritarianism, need for cognition, and social dominance (Author et al., 2017). There is no principled reason why the same approach could not be applied to virtually any latent trait for which a calibrated item bank can be assembled. Indeed, the combination of large publicly available item pools, increasingly accessible IRT calibration tools, and web-based survey infrastructure suggests that the present moment is a particularly opportune one for a broader transition toward adaptive personality assessment.

Future research should extend this framework to other constructs, examine its performance in clinical and applied selection contexts, and explore precision-based stopping rules that would allow the number of items to vary with the difficulty of achieving a reliable estimate for a given individual.

Beyond the BF-AI, the `catSurv` package already provides validated adaptive inventories for a range of constructs routinely measured in social and personality research, including Narcissism, Machiavellianism, Right-Wing Authoritarianism, Social Dominance Orientation, Need for Cognition, Need for Affect, Need to Evaluate, the Empathizing and Systemizing Quotients, and all four Schwartz Values dimensions (Montgomery & Rossiter, 2017). Each can be deployed through the same infrastructure used for the BF-AI, requiring no additional technical setup beyond a standard web-enabled survey platform.

4.3. Limitations

Several limitations of the present research warrant acknowledgment. First, the primary calibration sample—drawn from the myPersonality Facebook application—is a large but non-representative convenience sample. Although its extraordinary size provides stable parameter estimates, the demographic profile of Facebook app users in the calibration period may differ systematically from the populations in which the BF-AI would ultimately be deployed. The supplementary YouGov calibration sample was included precisely to address this concern, but future work should examine whether the item parameters, and thus the adaptive algorithm’s behavior, generalize across more

diverse populations including non-Western samples.

Second, the present study used a fixed-length termination rule of five items per trait. This was a deliberate methodological choice made to facilitate direct comparisons with fixed-length benchmarks, but it forgoes one of the most compelling advantages of conventional CAT designs: the ability to tailor test length to the difficulty of achieving a reliable estimate for a given individual. A precision-based stopping rule, under which administration continues until the standard error of the trait estimate falls below a predetermined threshold, would be expected to produce even more favorable results, particularly for individuals at extreme trait levels where more items may be needed to achieve stable estimates.

Third, the criterion validity analysis relied exclusively on self-report measures of life outcomes. While self-report is standard in this literature and the outcomes were drawn from a well-validated replication framework (Soto, 2019), demonstrating that the BF-AI predicts behavioral or informant-rated criteria would provide a stronger test of its validity. Similarly, the present study did not examine test-retest reliability, which is an important psychometric property for any measure intended for repeated deployment in longitudinal designs.

Finally, while the present paper focuses on measurement precision and criterion validity, it does not speak directly to questions of convergent and discriminant validity vis-à-vis other Big Five instruments. Future research should establish the degree to which BF-AI trait estimates converge with those produced by established full-length measures and examine whether the adaptive algorithm introduces any systematic biases in the measurement of specific facets.

With respect to the limits of generalizability, the present findings should be interpreted with some caution outside the contexts studied here. The criterion validity results are based on a U.S. online convenience sample (Lucid), and we would not necessarily expect the observed effect sizes to replicate precisely in non-WEIRD populations, in samples with markedly different demographic compositions, or for life outcomes whose relationship with personality is known to vary cross-culturally. Similarly, the simulation results assume a uniform trait distribution and a fixed five-item stopping rule; precision advantages may differ under alternative stopping criteria or in populations

where trait scores cluster near the mean. The adaptive algorithm's behavior also depends on the stability of the underlying IRT parameters, which were estimated primarily from English-speaking samples; applications to translated versions of the IPIP-NEO or to populations with very different item response patterns would require independent calibration.

4.4. Conclusions

Personality researchers have long recognized the psychometric costs of abbreviated assessment while simultaneously relying on short measures out of practical necessity. The present findings suggest that this compromise is no longer necessary. By leveraging adaptive item selection and a large, well-calibrated item bank, the BF-AI achieves measurement precision and criterion validity that meet or exceed those of fixed-length short forms while requiring no additional participant burden. The digital infrastructure to support this approach is already in place across virtually all contemporary data collection platforms. We hope the present research, together with the openly available tools provided through the `catSurv` package, encourages broader adoption of adaptive methods in personality research and beyond.

Open Practices

The analysis code and the cleaned data for Samples 2 and 3 will be made publicly available at the Harvard Dataverse upon publication. Sample 1 (the myPersonality Facebook dataset) cannot be shared: the data are not ours to redistribute, access is restricted by the original data custodians, and the terms of our IRB approval do not permit secondary distribution. The item parameters derived from Sample 1 are, however, publicly available as part of the open-source `catSurv` R package (Montgomery & Rossiter, 2017). This article has earned the Open Materials badge; the Open Data badge applies to Samples 2 and 3 only.

Author Contributions

Removed for blind review.

Declaration of Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work the authors used a generative AI tool in order to assist with editing, formatting, and organizing and documenting the replication archive. No complete sections of the manuscript were drafted using this tool. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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A. Life Outcome Measures: Complete Item Text

This appendix provides complete item wording, display conditions, response options, and scoring rules for all seven life outcome measures used in the criterion validity study. Items marked (R) were reverse-scored before computing composites. Six of the seven measures were implemented directly from the LOOPR project (Soto, 2019); Self-Rated Health is a standard single-item measure not part of the LOOPR battery.

1. Self-Rated Health. *Source:* Standard single-item self-rated health measure (Idler & Benyamini, 1997).

1. In general, would you say your health is ... (*1 = Poor, 2 = Fair, 3 = Good, 4 = Very good, 5 = Excellent*)

Scoring: Single item; used as-is.

2. Peer Acceptance. *Source:* LOOPR Peer & family relations: Peers' acceptance and friendship (Soto, 2019).

1. In relation to people you know, how would you describe your popularity among your peers? (*1 = Extremely unpopular, 2 = Very unpopular, 3 = Moderately unpopular, 4 = Somewhat unpopular, 5 = Average, 6 = Somewhat popular, 7 = Moderately popular, 8 = Very popular, 9 = Extremely popular*)

Scoring: Single item; used as-is.

3. Romantic Relationship Satisfaction. *Source:* LOOPR Romantic relations: Satisfaction (Soto, 2019).

Routing: Items 3–8 (RomanticSatisfaction_1–6) were displayed only to participants who reported being currently married or currently having a romantic partner of at least one month's duration; participants meeting neither condition were excluded from analyses involving this outcome.

Prompt (items 3–8): "The following questions ask about certain aspects of your relationship with your spouse or romantic partner. Please answer these questions for the present time in your relationship."

1. Are you currently married? (*1 = Yes, 2 = No*)
2. Do you currently have a romantic partner with whom you have been involved for at least one month? (*1 = Yes, 2 = No*)
3. To what extent do you reveal or disclose very intimate facts about yourself to your spouse or partner?
4. How close do you feel toward your spouse or partner?
5. To what extent do you have a sense of "belonging" with your spouse or partner?
6. How often do you and your spouse or partner argue with one another? (R)

7. To what extent do you feel that your spouse or partner demands or requires too much of your time and attention? (R)
8. To what extent do you try to change things about your spouse or partner that bother you? (R)

Response options (items 3–8): 1 = Not at all, 2, 3, 4, 5 = Very much.

Scoring: Mean of items 3–8; items 6–8 reverse-scored before averaging.

4. Job Satisfaction. *Source:* Minnesota Satisfaction Questionnaire (MSQ; Weiss & Vale, 1987), as implemented in LOOPR (Soto, 2019).

Display condition: Shown only to participants whose main current activity was “Employed.”

Prompt: “Below you will find statements about your present job. Decide *how satisfied you feel about the aspect of your job* described by the statement.”

1. The working conditions
2. The way my co-workers get along with each other
3. Being able to keep busy all the time
4. The chance to work alone on the job
5. The way my boss handles his/her workers
6. The competence of my supervisor in making decisions

Response options: 1 = Very dissatisfied, 2 = Dissatisfied, 3 = Neither satisfied nor dissatisfied, 4 = Satisfied, 5 = Very satisfied.

Scoring: Mean of items 1–6.

5. Volunteerism. *Source:* LOOPR (Soto, 2019).

1. Have you ever volunteered?
2. Are you currently volunteering?
3. Do you plan on volunteering during the next two months?

Response options: 0 = No, 1 = Yes.

Scoring: Mean of items 1–3.

6. Criminal Behavior. *Source:* LOOPR Criminality: Criminal behavior (Soto, 2019).

Prompt: “In your life, have you ever taken part in. . .”

1. Theft of something worth \$50 or less?
2. Theft of something worth more than \$50?
3. Vandalism?
4. Robbery?

5. Assault (with or without a weapon)?
6. Driving while intoxicated?

Response options: 1 = Never, 2 = Once or twice, 3 = Several times, 4 = Many times.

Scoring: Mean of items 1–6.

7. Religious Beliefs. *Source:* LOOPR Spirituality battery, items 25–30 (Religious beliefs and behavior subscale) (Soto, 2019).

Prompt: “Using the five-point scale below, rate the extent to which you agree with each statement as it applies to you.”

1. I believe that God or a Higher Power is responsible for my existence.
2. I believe that going to religious services is important.
3. I see God or a Higher Power present in all the things I do.
4. I seldom, if ever, pray. (R)
5. I have made no efforts to include religious practices as part of my life. (R)
6. I do not see a Higher Power having any influence over my life. (R)

Response options: 1 = Strongly disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly agree.

Scoring: Mean of items 1–6; items 4–6 reverse-scored before averaging so that higher scores indicate greater religiosity.

B. Sample Characteristics

Table 2 reports the demographic composition of Sample 3 ($N = 3,029$) after exclusions.

Table 2: Sample 3 Demographic Characteristics ($N = 3,029$)		
Characteristic	%	n
<i>Age</i>		
18-29	15.5	469
30-39	24.3	736
40-49	15.6	473
50-59	12.8	388
60-69	17.3	524
70-79	13	394
80+	1.6	48
<i>Gender</i>		
Female	52.4	1587
Male	47.3	1433
Other	0.2	6
<i>Race/Ethnicity</i>		
Asian	9.3	282
Black or African American	13.1	397
Hispanic/Latinx	10.7	324
Multiracial/Other	9.9	300
Native American	1.1	33
Other/Prefer not to say	1.2	36
Pacific Islander	0.2	6
White	54.6	1654
<i>Education</i>		
Associate's degree	10.7	324
Bachelor's degree	28.2	854
Doctorate degree (For example: PhD, EdD)	3	91
High school diploma or GED	17.3	524
Master's degree (For example: MA, MBA, MSW)	16.8	509
Professional school degree (For example: MD, JD, DDS)	1.7	51
Some college	20.5	621
Some high school	1.8	55

Note. Percentages may not sum to 100 due to rounding or missing responses.

C. Simulation Procedure

This appendix describes the step-by-step procedure used to evaluate measurement precision in Study 1. The procedure proceeds in two stages: first, harmonizing item parameters from multiple inventories onto a common latent scale; second, simulating evaluation data and estimating traits under each comparison condition.

Stage 1: Item Parameter Harmonization

Step 1. Estimate initial GRM parameters. A Graded Response Model (GRM) was fitted separately to each inventory using its respective calibration dataset, as described in the main text: the IPIP-NEO parameters were taken from Sample 2 described in the main text; the BFI-2 and BFI-2-XS were calibrated from the LOOPR dataset (Soto, 2019); and the TIPI, IPIP-50, and Mini-IPIP were calibrated from the Lucid validation sample (Sample 3). Because the IPIP-50 is a 50-item subset of the IPIP-NEO and the BFI-2-XS is a 15-item *subset* of the BFI-2, neither required separate calibration. At this stage, item parameters across inventories are expressed in incomparable metrics.

Step 2. Generate a bridge sample. Draw $N = 1,000$ simulees with true trait scores $\theta_i \stackrel{\text{iid}}{\sim} \text{Unif}[-4, 4]$ for each of the five Big Five traits independently. The $[-4, 4]$ range ensures that all ordered response categories are observed even for items with extreme thresholds.

Step 3. Simulate bridge responses using the GRM. For each simulee i and item j with $K_j + 1$ ordered response categories ($k = 0, 1, \dots, K_j$), the cumulative probability of responding in category k or higher is:

$$P^*(X_{ij} \geq k \mid \theta_i) = \frac{\exp[\alpha_j(\theta_i - \beta_{jk})]}{1 + \exp[\alpha_j(\theta_i - \beta_{jk})]} \quad (1)$$

where $\alpha_j > 0$ is the item discrimination parameter and $\beta_{j1} < \beta_{j2} < \dots < \beta_{jK_j}$ are the ordered threshold parameters. The probability of a response in exactly category k is then:

$$P(X_{ij} = k \mid \theta_i) = P^*(X_{ij} \geq k \mid \theta_i) - P^*(X_{ij} \geq k + 1 \mid \theta_i) \quad (2)$$

with boundary conditions $P^*(X_{ij} \geq 0) \equiv 1$ and $P^*(X_{ij} \geq K_j + 1) \equiv 0$. Responses are drawn from this categorical distribution and applied to all items from all inventories for each bridge simulee.

Step 4. Re-estimate parameters jointly. GRM parameters are re-estimated from the bridge sample responses in a single joint model per trait. Because all bridge simulees share a common θ distribution, this joint re-estimation places every item—regardless of its origin inventory—onto a single common latent scale. The resulting harmonized parameters $\hat{\alpha}_j$ and $\hat{\beta}_{jk}$ are used in all subsequent steps.

Stage 2: Precision Evaluation

Step 5. Generate an evaluation sample. Draw a fresh $N = 1,000$ simulees with $\theta_i \stackrel{\text{iid}}{\sim} \text{Unif}[-4, 4]$ per trait. These true θ values serve as ground truth throughout. All evaluation conditions within a trait use the same 1,000 simulees.

Step 6. Pre-simulate complete response records. For each evaluation simulee i , apply Equations 1–2 with the harmonized parameters from Step 4 to generate a simulated response to *every* item in the unified bank. This yields one complete response vector per simulee, fixed before any estimation is performed. The three estimation conditions in Step 7 all read from this same pre-computed record; no responses are re-drawn during estimation.

Step 7. Estimate traits under three conditions. For each trait, test length $\ell \in \{2, 3, 4, 5, 6, 7\}$ items per trait, and simulee, three estimation conditions are applied. All three read from the pre-simulated response record in Step 6, differing only in which items they access:

- (a) *BF-AI (adaptive)*. Items are selected sequentially. At step t , the algorithm reads the simulee’s pre-simulated response to the chosen item and updates the trait estimate using the CAT as described. Because item selection at each step depends on all prior responses, each simulee follows a unique item sequence.
- (b) *Fixed benchmark*. Read the simulee’s pre-simulated responses to the pre-designated items for the relevant inventory at that length—TIPI (2 items/trait), BFI-2-XS (3), Mini-IPIP (4), IPIP-50 (5), BFI-2 (6), IPIP-NEO (7)—and compute an EAP estimate (Equation ??) from that fixed response subset. All simulees access the same items.
- (c) *Random baseline*. For each simulee independently, draw a random subset of ℓ items without replacement from the full unified bank; read the pre-simulated responses to those items; compute an EAP estimate. Provides a lower-bound precision reference for each test length.

Step 8. Compute absolute bias. For each simulee i , trait, length, and condition, compute $|\theta_i - \hat{\theta}_i|$.

Step 9. Smooth over the θ distribution. Absolute bias is summarized as a smooth function of true θ using a, estimated separately per trait, length, and condition. Smoothed curves appear in Figures 1 and 2; full numerical means and standard deviations are in Table 3 (Appendix D).

Implementation. All steps were implemented in R using the `catSurv` package (Author et al., 2017). The 30 evaluation conditions (5 traits $\times$ 6 lengths) were executed in parallel across 10 workers. See `02_run_simulation.R` in the replication archive.

D. Simulation Bias: Full Numerical Results

Table 3 reports mean absolute bias (M) and standard deviation (SD) for each method, test length, and Big Five trait in the simulation study. Fixed inventories at each length are: two items = TIPI; three items = BFI-2-XS; four items = Mini-IPIP; five items = IPIP-50; six items = BFI-2; seven items = IPIP-NEO.

Table 3: Mean Absolute Bias by Method, Test Length, and Trait

Items	Trait	CAT		Fixed		Random	
		<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
2	Agreeableness	0.743	0.619	0.942	0.713	0.996	0.749
	Conscientiousness	0.732	0.608	0.86	0.692	0.943	0.7
	Extraversion	0.78	0.682	0.902	0.774	0.97	0.715
	Neuroticism	0.83	0.686	0.889	0.741	0.957	0.739
	Openness	0.889	0.709	1.057	0.797	1.105	0.761
3	Agreeableness	0.595	0.516	0.862	0.713	0.849	0.693
	Conscientiousness	0.578	0.499	0.856	0.716	0.811	0.625
	Extraversion	0.667	0.602	0.87	0.64	0.843	0.653
	Neuroticism	0.69	0.601	0.873	0.735	0.842	0.671
	Openness	0.749	0.613	0.95	0.755	0.929	0.689
4	Agreeableness	0.519	0.463	0.828	0.713	0.766	0.604
	Conscientiousness	0.503	0.441	0.776	0.624	0.712	0.587
	Extraversion	0.584	0.538	0.742	0.638	0.748	0.601
	Neuroticism	0.603	0.538	0.776	0.654	0.761	0.65
	Openness	0.628	0.52	0.827	0.651	0.831	0.631
5	Agreeableness	0.466	0.424	0.609	0.567	0.689	0.566
	Conscientiousness	0.448	0.403	0.53	0.471	0.636	0.532
	Extraversion	0.512	0.464	0.566	0.532	0.684	0.568
	Neuroticism	0.542	0.502	0.564	0.532	0.71	0.617
	Openness	0.555	0.458	0.556	0.487	0.735	0.581
6	Agreeableness	0.432	0.392	0.541	0.544	0.624	0.542
	Conscientiousness	0.417	0.373	0.535	0.551	0.609	0.514
	Extraversion	0.471	0.433	0.483	0.398	0.647	0.535
	Neuroticism	0.502	0.477	0.54	0.514	0.676	0.593
	Openness	0.499	0.416	0.577	0.561	0.686	0.554
7	Agreeableness	0.401	0.368	0.385	0.375	0.588	0.512
	Conscientiousness	0.395	0.352	0.373	0.376	0.565	0.489
	Extraversion	0.438	0.406	0.369	0.346	0.598	0.518
	Neuroticism	0.469	0.454	0.416	0.436	0.631	0.567
	Openness	0.458	0.384	0.417	0.393	0.648	0.522

Note. Based on 1,000 simulated respondents drawn from $\text{Unif}[-4, 4]$. CAT = adaptive item selection; Fixed benchmarks by length: two items = TIPI, three items = BFI-2-XS, four items = Mini-IPIP, five items = IPIP-50 (10 items/trait), six items = BFI-2 (15 items/trait), seven items = IPIP-NEO (20 items/trait); Random = random item selection.

E. Full Criterion Validity Results

Tables 4–8 report Pearson correlations (r) with 95% confidence intervals for every trait $\times$ outcome $\times$ method combination, presented separately for each Big Five trait. CAT-2 through CAT-5 refer to the BF-AI administered with two through five items per trait. Confidence intervals appear in smaller text below each coefficient.

Table 4: Criterion Validity Correlations: Extraversion

Outcome	CAT-2	CAT-3	CAT-4	CAT-5	TIPI	BFI-XS	miniIPIP
Criminal Behavior	.02 [-.01, .06]	-.02 [-.06, .02]	-.04 [-.08, -.01]	-.02 [-.06, .01]	.00 [-.03, .04]	.05 [.02, .09]	.03 [-.00, .07]
Health	.31 [.28, .34]	.30 [.26, .33]	.29 [.25, .32]	.29 [.26, .32]	.20 [.16, .23]	.33 [.30, .36]	.26 [.23, .29]
Job Satisfaction	.31 [.26, .35]	.31 [.26, .35]	.30 [.26, .35]	.31 [.27, .35]	.13 [.08, .18]	.23 [.19, .28]	.23 [.18, .27]
Peer Acceptance	.45 [.42, .48]	.48 [.45, .50]	.48 [.45, .51]	.49 [.46, .52]	.34 [.31, .37]	.41 [.38, .44]	.46 [.43, .49]
Religious Beliefs	.14 [.11, .18]	.17 [.13, .20]	.18 [.15, .21]	.17 [.14, .21]	.11 [.07, .14]	.12 [.08, .15]	.10 [.06, .13]
Romantic Satisfaction	.07 [.03, .11]	.11 [.07, .15]	.12 [.08, .16]	.10 [.06, .15]	.08 [.04, .12]	.05 [.01, .09]	.04 [-.00, .08]
Volunteerism	.22 [.18, .25]	.21 [.17, .24]	.19 [.16, .23]	.20 [.16, .23]	.14 [.11, .18]	.20 [.17, .24]	.18 [.14, .21]

Note. Pearson r [95% CI], $N = 3,029$.

Table 5: Criterion Validity Correlations: Agreeableness

Outcome	CAT-2	CAT-3	CAT-4	CAT-5	TIPI	BFI-XS	miniIPIP
Criminal Behavior	-.24 [-.27, -.20]	-.25 [-.28, -.22]	-.25 [-.28, -.21]	-.24 [-.27, -.20]	-.27 [-.30, -.24]	-.22 [-.26, -.19]	-.22 [-.25, -.18]
Health	.13 [.09, .16]	.14 [.10, .17]	.16 [.13, .19]	.17 [.13, .20]	.07 [.04, .11]	.13 [.10, .17]	.09 [.06, .13]
Job Satisfaction	.26 [.21, .31]	.28 [.24, .33]	.32 [.28, .36]	.35 [.30, .39]	.22 [.17, .27]	.27 [.22, .31]	.27 [.22, .31]
Peer Acceptance	.22 [.18, .25]	.24 [.21, .28]	.27 [.24, .30]	.28 [.25, .31]	.17 [.14, .20]	.24 [.20, .27]	.24 [.21, .28]
Religious Beliefs	.18 [.15, .21]	.21 [.17, .24]	.22 [.19, .26]	.23 [.20, .27]	.25 [.22, .29]	.25 [.22, .28]	.28 [.24, .31]
Romantic Satisfaction	.21 [.17, .25]	.22 [.18, .26]	.23 [.19, .27]	.24 [.20, .28]	.32 [.28, .36]	.24 [.20, .28]	.22 [.18, .26]
Volunteerism	.10 [.07, .14]	.11 [.07, .14]	.13 [.09, .16]	.15 [.11, .18]	.05 [.01, .08]	.11 [.08, .15]	.15 [.12, .19]

Note. Pearson r [95% CI], $N = 3,029$.

Table 6: Criterion Validity Correlations: Conscientiousness

Outcome	CAT-2	CAT-3	CAT-4	CAT-5	TIPI	BFI-XS	miniIPIP
Criminal Behavior	-.15 [-.18, -.11]	-.15 [-.19, -.12]	-.18 [-.21, -.15]	-.17 [-.21, -.14]	-.26 [-.29, -.22]	-.27 [-.30, -.24]	-.22 [-.25, -.19]
Health	.27 [.24, .30]	.29 [.25, .32]	.29 [.26, .32]	.30 [.27, .33]	.21 [.17, .24]	.24 [.20, .27]	.21 [.17, .24]
Job Satisfaction	.28 [.24, .33]	.30 [.26, .35]	.31 [.26, .35]	.33 [.29, .38]	.25 [.21, .30]	.26 [.21, .30]	.22 [.17, .26]
Peer Acceptance	.25 [.22, .29]	.26 [.23, .29]	.26 [.23, .29]	.27 [.24, .31]	.24 [.21, .28]	.24 [.21, .27]	.18 [.14, .21]
Religious Beliefs	.16 [.13, .20]	.16 [.13, .20]	.18 [.14, .21]	.18 [.14, .21]	.21 [.17, .24]	.18 [.15, .21]	.19 [.15, .22]
Romantic Satisfaction	.23 [.19, .27]	.24 [.20, .28]	.26 [.22, .30]	.26 [.22, .30]	.30 [.27, .34]	.33 [.29, .37]	.30 [.26, .33]
Volunteerism	.10 [.07, .14]	.10 [.07, .14]	.10 [.06, .13]	.11 [.07, .14]	.05 [.01, .09]	.02 [-.01, .06]	.02 [-.01, .06]

Note. Pearson r [95% CI], $N = 3,029$.

Table 7: Criterion Validity Correlations: Neuroticism

Outcome	CAT-2	CAT-3	CAT-4	CAT-5	TIPI	BFI-XS	miniIPIP
Criminal Behavior	.16 [.13, .20]	.15 [.12, .19]	.14 [.10, .17]	.14 [.11, .18]	.15 [.11, .18]	.16 [.12, .19]	.14 [.11, .18]
Health	-.24 [-.28, -.21]	-.26 [-.29, -.23]	-.28 [-.32, -.25]	-.29 [-.32, -.25]	-.30 [-.33, -.27]	-.31 [-.34, -.28]	-.24 [-.27, -.21]
Job Satisfaction	-.22 [-.27, -.17]	-.26 [-.30, -.21]	-.27 [-.32, -.23]	-.28 [-.32, -.23]	-.30 [-.35, -.26]	-.28 [-.33, -.24]	-.25 [-.30, -.21]
Peer Acceptance	-.24 [-.27, -.21]	-.28 [-.31, -.24]	-.30 [-.33, -.26]	-.31 [-.35, -.28]	-.32 [-.35, -.29]	-.33 [-.36, -.29]	-.25 [-.28, -.22]
Religious Beliefs	-.16 [-.20, -.13]	-.18 [-.21, -.14]	-.17 [-.20, -.14]	-.17 [-.20, -.14]	-.17 [-.20, -.13]	-.16 [-.19, -.12]	-.15 [-.19, -.12]
Romantic Satisfaction	-.29 [-.33, -.25]	-.30 [-.34, -.26]	-.30 [-.33, -.26]	-.30 [-.34, -.26]	-.29 [-.33, -.25]	-.30 [-.34, -.26]	-.32 [-.36, -.28]
Volunteerism	-.05 [-.09, -.02]	-.06 [-.10, -.03]	-.07 [-.10, -.03]	-.07 [-.11, -.04]	-.07 [-.10, -.03]	-.06 [-.10, -.03]	-.08 [-.11, -.04]

Note. Pearson r [95% CI], $N = 3,029$.

Table 8: Criterion Validity Correlations: Openness

Outcome	CAT-2	CAT-3	CAT-4	CAT-5	TIPI	BFI-XS	miniIPIP
Criminal Behavior	-.07 [-.11, -.04]	-.09 [-.13, -.06]	-.06 [-.10, -.03]	-.05 [-.08, -.01]	-.13 [-.16, -.09]	-.05 [-.08, -.01]	-.12 [-.16, -.09]
Health	.10 [.06, .13]	.12 [.08, .15]	.13 [.09, .16]	.13 [.10, .17]	.17 [.13, .20]	.15 [.11, .18]	.09 [.06, .13]
Job Satisfaction	.19 [.14, .24]	.22 [.18, .27]	.25 [.20, .29]	.26 [.21, .30]	.22 [.17, .27]	.19 [.14, .24]	.18 [.13, .23]
Peer Acceptance	.13 [.10, .17]	.17 [.13, .20]	.18 [.14, .21]	.19 [.16, .23]	.27 [.24, .30]	.22 [.18, .25]	.17 [.14, .21]
Religious Beliefs	.11 [.07, .14]	.13 [.10, .17]	.13 [.09, .16]	.13 [.10, .17]	.15 [.11, .18]	.13 [.09, .16]	.14 [.11, .18]
Romantic Satisfaction	.12 [.07, .16]	.15 [.11, .19]	.15 [.11, .19]	.14 [.10, .19]	.17 [.13, .21]	.14 [.10, .18]	.18 [.14, .22]
Volunteerism	.13 [.10, .17]	.14 [.11, .18]	.15 [.11, .18]	.17 [.13, .20]	.12 [.09, .16]	.16 [.12, .19]	.09 [.05, .12]

Note. Pearson r [95% CI], $N = 3,029$.

F. Most Frequently Selected Items by Trait

Table 9 shows the eight most frequently selected items per Big Five trait across Sample 3 participants under the five-item adaptive battery, providing a descriptive picture of which items the adaptive algorithm most commonly administers in practice.

Table 9: Most Frequently Selected IPIP-NEO Items per Trait in Sample 3 (Five-Item Battery)

Trait	Item	Item Text (stem: "I...")	Selections	%
<i>Extraversion</i>				
	q83	Feel comfortable around people.	3029	100%
	q69	Find it difficult to approach others.	2792	92.2%
	q93	Start conversations.	2738	90.4%
	q43	Talk to a lot of different people at parties.	2344	77.4%
	q10	Make friends easily.	2184	72.1%
	q63	Am the life of the party.	884	29.2%
	q73	Am skilled in handling social situations.	397	13.1%
	q14	Keep in the background.	378	12.5%
<i>Agreeableness</i>				
	q92	Insult people.	3029	100%
	q6	Believe that others have good intentions.	2945	97.2%
	q86	Have a good word for everyone.	2799	92.4%
	q9	Cut others to pieces.	2497	82.4%
	q66	Respect others.	1543	50.9%
	q56	Trust what people say.	741	24.5%
	q32	Get back at others.	727	24%
	q52	Contradict others.	426	14.1%
<i>Conscientiousness</i>				
	q65	Finish what I start.	3029	100%
	q95	Carry out my plans.	2417	79.8%
	q55	Make plans and stick to them.	2241	74%
	q35	Follow through with my plans.	2075	68.5%
	q58	Leave things unfinished.	1999	66%
	q28	Waste my time.	1298	42.9%
	q75	Get chores done right away.	785	25.9%
	q8	Need a push to get started.	461	15.2%
<i>Neuroticism</i>				
	q80	Am often down in the dumps.	3029	100%
	q17	Get stressed out easily.	2930	96.7%
	q60	Worry about things.	2656	87.7%
	q50	Panic easily.	2202	72.7%
	q47	Am not easily frustrated.	2197	72.5%
	q90	Am filled with doubts about things.	826	27.3%
	q30	Dislike myself.	810	26.7%
	q37	Seldom feel blue.	232	7.7%
<i>Openness</i>				
	q64	Have difficulty understanding abstract ideas.	3030	100%
	q24	Am not interested in abstract ideas.	3029	100%
	q91	Believe in the importance of art.	3007	99.3%
	q54	Am not interested in theoretical discussions.	2960	97.7%
	q74	Do not like art.	2563	84.6%
	q34	Do not enjoy going to art museums.	399	13.2%
	q61	Enjoy hearing new ideas.	68	2.2%
	q7	Avoid philosophical discussions.	67	2.2%

Note. Counts reflect item selections across all five administered positions of the BF-AI five-item adaptive battery ($N = 3,029$). An item selected at every position for every respondent would have a count of 3,029 (100%). Full item wording uses the stem "I..."; see Goldberg et al. (2006) for the complete IPIP-NEO item pool.

G. Convergent Validity with Fixed Short Forms

Figure 5 displays the full pairwise correlation matrices between all seven measures (TIPI, BFI-XS, miniIPIP, and BF-AI at 2–5 items) for each of the five Big Five traits, computed within Sample 3. Each cell shows the Pearson r ; darker shading indicates higher convergence. Table 10 reproduces the BF-AI-to-fixed comparisons at matched lengths in tabular form.

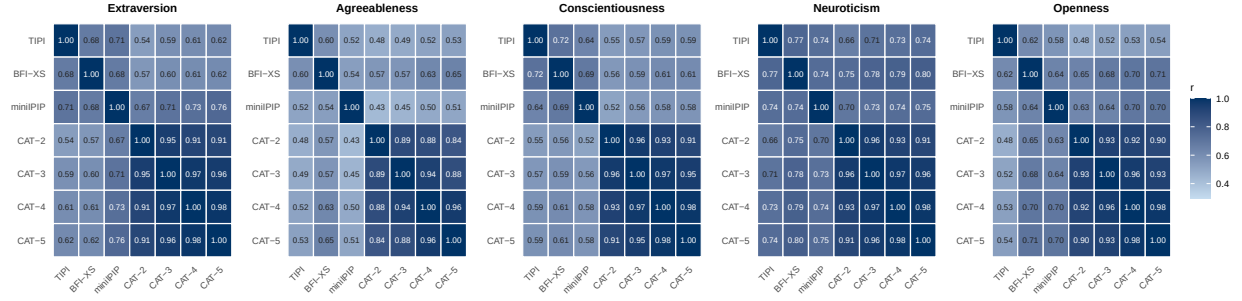


Figure 5: Convergent Validity: Pairwise Correlations Among All Short-Form Measures

Note. Pearson correlations computed within Sample 3 ($N = 3,029$). Darker blue indicates higher convergence. The color scale runs from $r = .30$ (light) to $r = 1.00$ (dark). BF-AI scores are EAP $\hat{\theta}$ estimates; fixed short form scores are summed/averaged item responses. CAT-2 through CAT-5 refer to the BF-AI administered with two through five items per trait.

Table 10: Convergent Validity: Correlations Between BF-AI Estimates and Fixed Short Form Scores

Trait	CAT-2 vs. TIPI	CAT-3 vs. BFI-XS	CAT-4 vs. miniPIP	CAT-5 vs. TIPI	CAT-5 vs. BFI-XS
Extraversion	0.54	0.60	0.73	0.62	0.62
Agreeableness	0.48	0.57	0.50	0.53	0.65
Conscientiousness	0.55	0.59	0.58	0.59	0.61
Neuroticism	0.66	0.78	0.74	0.74	0.80
Openness	0.48	0.68	0.70	0.54	0.71

Note. All correlations computed within Sample 3 ($N = 3,029$). BF-AI scores are EAP theta estimates; fixed short form scores are summed/averaged item responses.

H. GAM Robustness Check

Pearson correlations assume linear relationships between personality traits and outcomes. To verify this assumption, we fitted both a Generalized Additive Model (GAM; with $k = 4$ smoothing splines, allowing flexible nonlinear curves) and a standard linear model to each of the 35 trait $\times$ outcome pairs using CAT-5 trait estimates. Table 11 reports R^2 and Akaike Information Criterion (AIC) for both specifications. A difference in AIC greater than 2 in favor of the GAM would indicate meaningful nonlinearity warranting concern. In practice, the linear and GAM fits are nearly indistinguishable across all 35 pairs: no trait–outcome combination shows a $\Delta\text{AIC} > 2$, and R^2 values are virtually identical between specifications. The personality–outcome relationships examined here are well-characterized as linear, and Pearson r is an appropriate and unbiased summary of these associations.

Table 11: GAM vs. Linear Model Fit for Criterion Validity Associations (CAT-5)

Trait	Outcome	R^2 (GAM)	R^2 (Linear)	Δ AIC	Preferred
Extraversion	Criminal Behavior	0	0	0	Linear
	Health	0.088	0.084	9.6	GAM
	Job Satisfaction	0.113	0.096	29.3	GAM
	Peer Acceptance	0.244	0.241	9.9	GAM
	Religious Beliefs	0.03	0.03	0	Linear
	Romantic Satisfaction	0.011	0.01	0.6	Linear
	Volunteerism	0.038	0.038	0.4	Linear
Agreeableness	Criminal Behavior	0.071	0.056	44.8	GAM
	Health	0.036	0.028	24.5	GAM
	Job Satisfaction	0.133	0.12	20.6	GAM
	Peer Acceptance	0.078	0.078	0	Linear
	Religious Beliefs	0.057	0.054	8.5	GAM
	Romantic Satisfaction	0.064	0.058	11.7	GAM
	Volunteerism	0.023	0.022	1.2	Linear
Conscientiousness	Criminal Behavior	0.037	0.029	21.1	GAM
	Health	0.097	0.09	19.5	GAM
	Job Satisfaction	0.131	0.111	33.6	GAM
	Peer Acceptance	0.082	0.075	20.3	GAM
	Religious Beliefs	0.032	0.032	0	Linear
	Romantic Satisfaction	0.066	0.066	0.2	Linear
	Volunteerism	0.015	0.012	7.4	GAM
Neuroticism	Criminal Behavior	0.022	0.02	5.4	GAM
	Health	0.083	0.082	1.9	Linear
	Job Satisfaction	0.099	0.077	37.4	GAM
	Peer Acceptance	0.102	0.098	12.1	GAM
	Religious Beliefs	0.029	0.029	0	Linear
	Romantic Satisfaction	0.089	0.089	0	Linear
	Volunteerism	0.006	0.005	2.2	GAM
Openness	Criminal Behavior	0.005	0.002	8.3	GAM
	Health	0.017	0.017	0	Linear
	Job Satisfaction	0.082	0.067	24.6	GAM
	Peer Acceptance	0.037	0.037	0	Linear
	Religious Beliefs	0.017	0.017	0	Linear
	Romantic Satisfaction	0.02	0.02	0	Linear
	Volunteerism	0.027	0.027	0.1	Linear

Note. Δ AIC = $AIC_{\text{linear}} - AIC_{\text{GAM}}$; positive values favor the GAM. Values > 2 suggest meaningful nonlinearity.

I. Criterion Validity by Gender

To assess whether the BF-AI's criterion validity generalizes across demographic groups, Table 12 and Figure 6 replicate the main criterion validity analysis (CAT-5) separately for men and women. Across all five traits and seven outcomes, correlations are broadly consistent across gender groups, with no systematic pattern of attenuation or enhancement for either group, suggesting the BF-AI performs equitably regardless of respondent gender.

Table 12: CAT-5 Criterion Validity: Pearson Correlations Between BF-AI Trait Estimates and Seven Life Outcomes, Separately for Men and Women

Trait	Outcome	Female	Male
Extraversion	Health	.26	.30
	Religious Beliefs	.18	.19
	Peer Acceptance	.50	.47
	Romantic Satisfaction	.08	.13
	Job Satisfaction	.27	.34
	Volunteerism	.19	.19
	Criminal Behavior	-.02	-.06
Agreeableness	Health	.15	.20
	Religious Beliefs	.23	.24
	Peer Acceptance	.26	.31
	Romantic Satisfaction	.23	.25
	Job Satisfaction	.36	.33
	Volunteerism	.15	.15
	Criminal Behavior	-.23	-.25
Conscientiousness	Health	.31	.28
	Religious Beliefs	.21	.16
	Peer Acceptance	.25	.29
	Romantic Satisfaction	.24	.28
	Job Satisfaction	.32	.34
	Volunteerism	.06	.15
	Criminal Behavior	-.16	-.21
Neuroticism	Health	-.32	-.22
	Religious Beliefs	-.21	-.16
	Peer Acceptance	-.33	-.25
	Romantic Satisfaction	-.26	-.36
	Job Satisfaction	-.25	-.29
	Volunteerism	-.08	-.05
	Criminal Behavior	.14	.21
Openness	Health	.10	.17
	Religious Beliefs	.12	.15
	Peer Acceptance	.17	.22
	Romantic Satisfaction	.16	.13
	Job Satisfaction	.28	.24
	Volunteerism	.18	.14
	Criminal Behavior	.01	-.10

Note. Correlations are computed separately within each gender group using CAT-5 theta estimates in Sample 3. Only men and women are shown; other gender categories had fewer than 200 respondents and are omitted. Correlations are broadly consistent across groups.

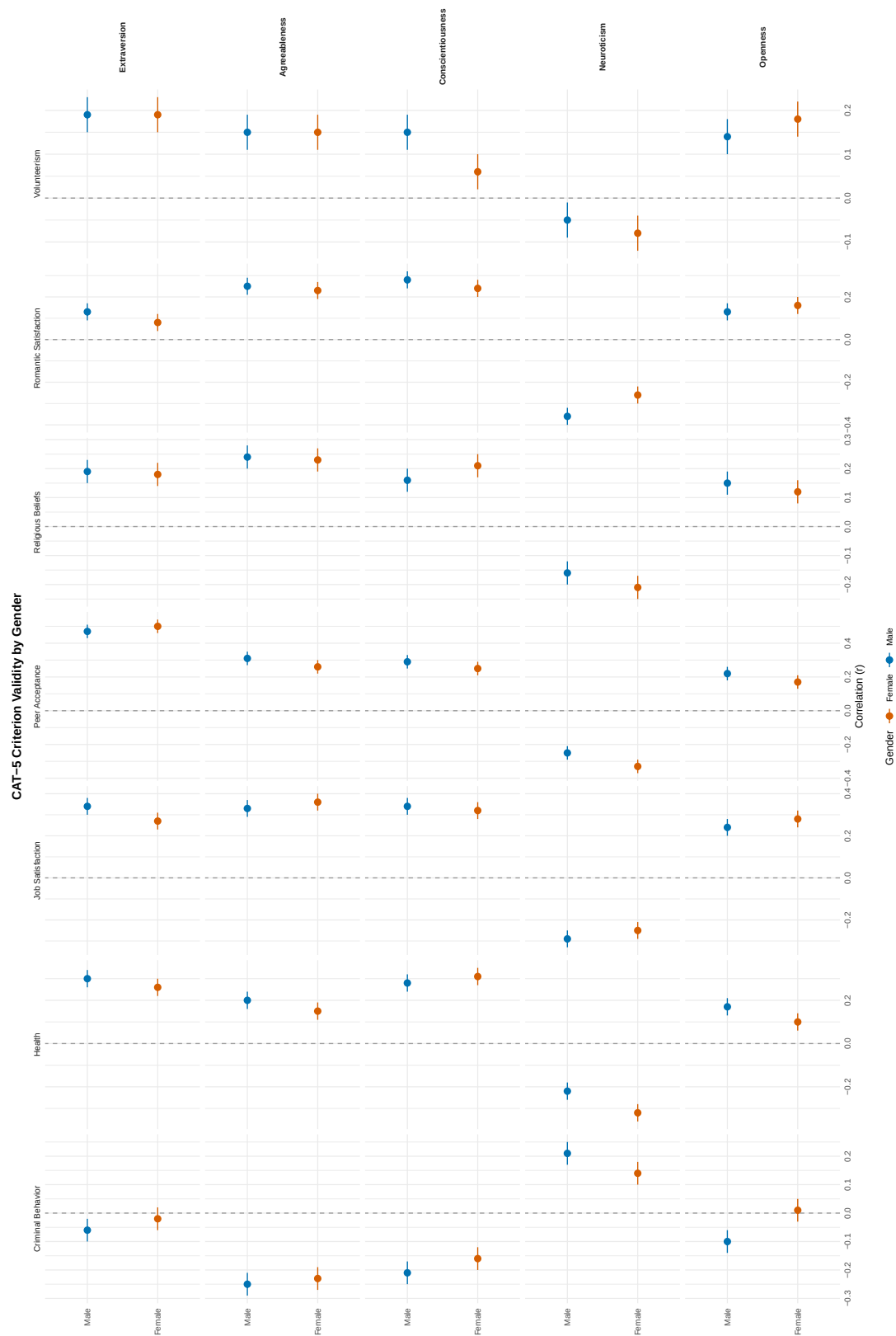


Figure 6: CAT-5 Criterion Validity by Gender

Note. Correlations between CAT-5 trait estimates and each life outcome, shown separately for men and women. Error bars are approximate 95% confidence intervals. Correlations are broadly consistent across gender groups with no systematic pattern of attenuation or enhancement.